Activity: Arduino Beat Box

Purpose: To make an arduino code that uses a photoresistor to make different musical notes. Because a photoresistor is a variable resistor, we can change the frequencies (musical notes) by moving our hands over the photoresistor(s) to adjust the amount of light the photoresistor receives, thus changing the frequencies of the notes based on the arduino code. This program will play three different tones.

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 3 - 220 Ω Resistors, 1- 10K Ω Resistor, 1 piezo speaker, 1 potentiometer.

NOTE: For an extension we can add multiple photoresistors to add multiple tones. For example, if you wanted two photoresistors, just double the materials needed. You will still only need one Arduino board, and one breadboard.

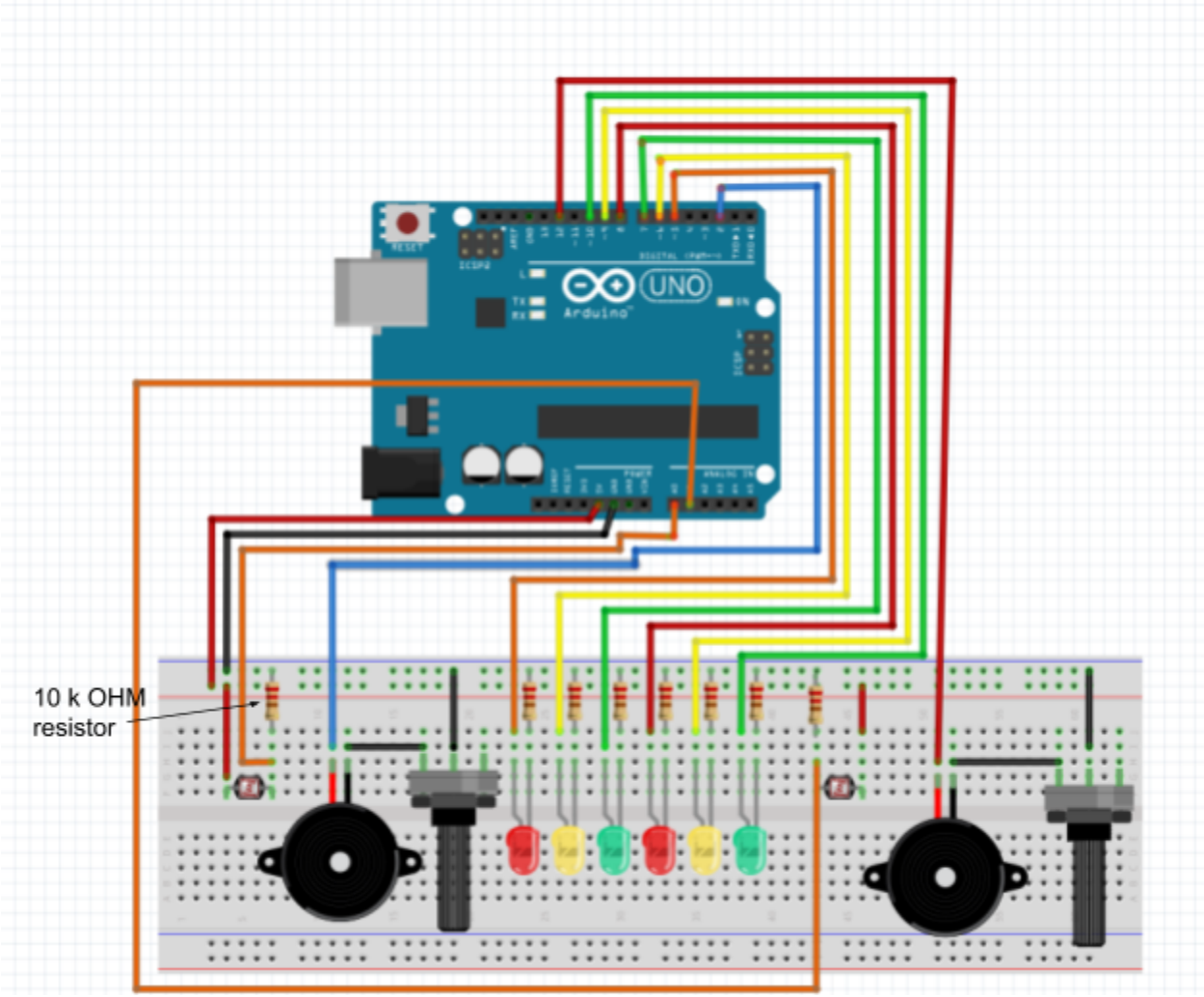
New Components:



Photoresistor. Photoresistors are light sensitive, variable resistors. As more light shines on the sensor's head, the resistance between its two terminals decreases.
Therefore photoresistors can act as an switch that uses light detection to turn loads (such as light) on and off or in this case, make some funky tones.

Procedure:

1. Wire the project using the following schematic.



2. You can create your code in the BlocklyDuino Editor then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

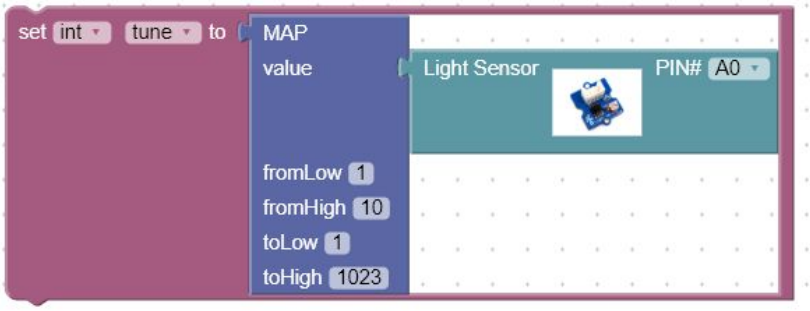
3. Once in BlocklyDuino, simply “drag” code blocks to create your code. Follow the steps below to make your code. There is also a video link that you can click on to assist you with your code.
VIDEO: ARDUINO BEATBOX

4. A. First we need set up a variable. These are similar to X and Y you use in math class. **Variables** are used to store information to be referenced and manipulated in a computer program. We will create the integer variable “tune”. You name your variable by using the drop down variable.

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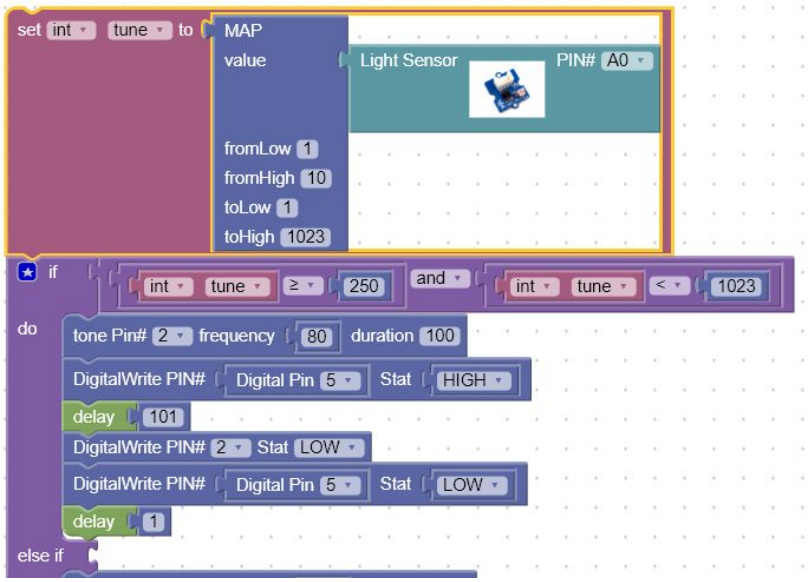
B. We are also going to use the "**map**" function. It re-maps a number from one range to another. That is, a value of fromLow would get mapped to toLow, a value of fromHigh to toHigh, values in-between to values in-between, etc. For this program, we are going to map values from 1 to 1024 and give them a range between 1 and 10 for coding purposes. These are the values we are going to use for the rest of our code to tell the arduino what to do with the amount of light the photoresistor receives.



C. In this program, we are going to be able to play 3 tones depending on how much light the light sensor receives. Therefore we are going to use a logic block. We are going to state that if the light received is between 250 and 1023, the speaker will play a frequency of 80 hz. (musical note). You will need to get the "and" logic block that allows you to create your range. It will also light the red LED which is connected to pin 5. At this point, your code should look like below.

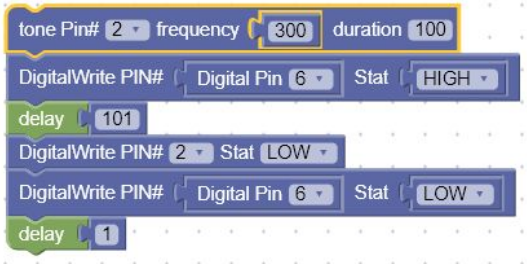


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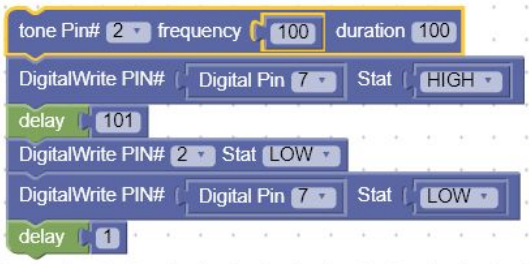


D. We are now going to complete the rest of the code by using the following blocks and placing them into the logic else if and do blocks. Remember to duplicate blocks and then just change the values to make your life easier. Good luck.









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*****SAVE your sketch to your BlocklyDuino Folder that you created*****

E. Once you are done putting the blocks in the correct order, click on the Arduino button to convert the blocks into Arduino code.



F. Upload your program. Observe your LED and complete your worksheet titled Assignment: Arduino Beat Box.

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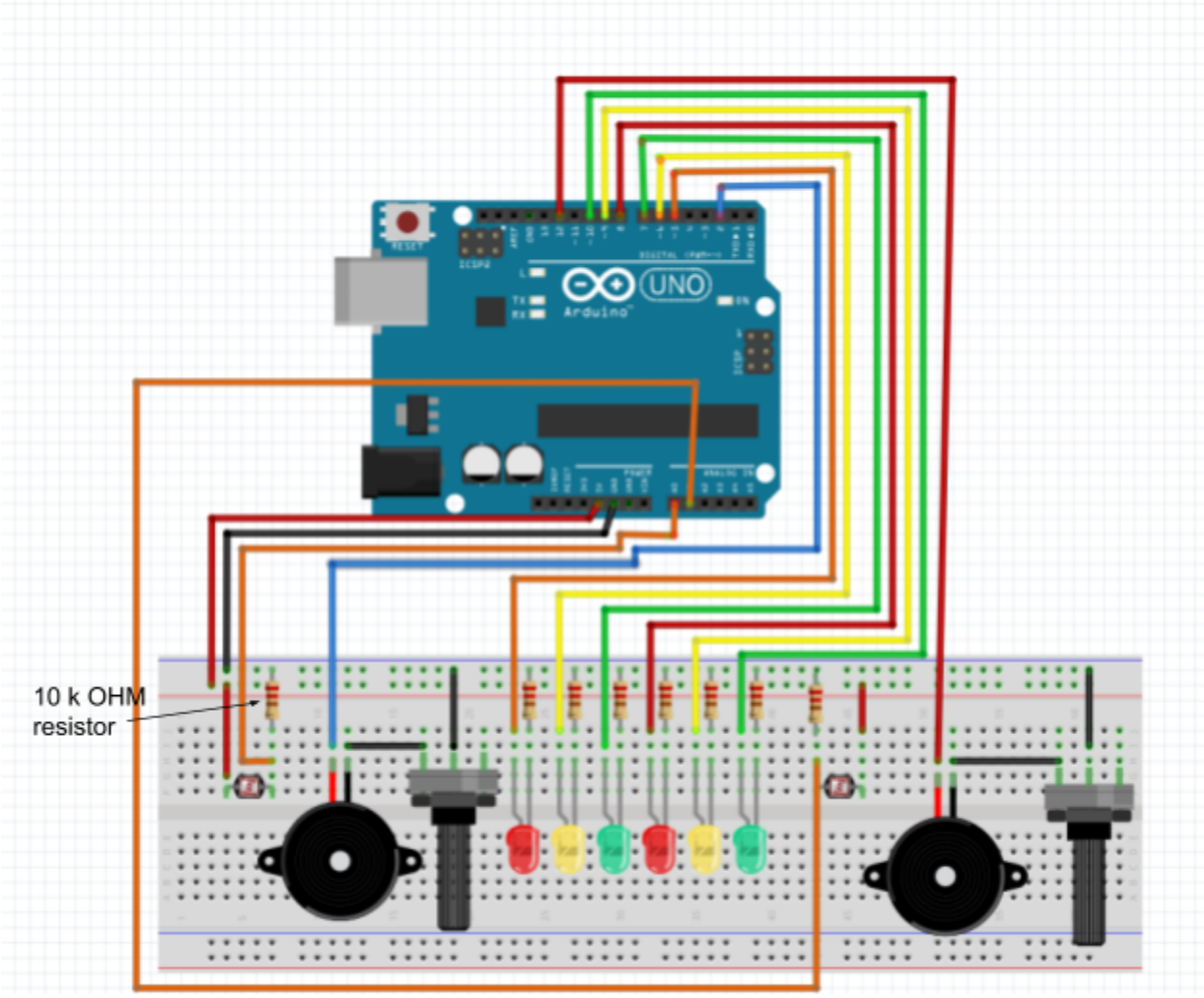
Assignment Worksheet: Arduino Beat Box

1. Move you hand over the photoresistor, what happens when you your hand is closer to the photoresistor?

2. Change the frequencies in the code to find the tones that best appeal to you. Reload your program to experiment with different tones.

Extension Challenges:

2. Add another set of lights, another resistor, and another speaker to create multiple sounds. Duplicate the code, be careful to switch the pins. Below is a schematic that you could use. You could even add more sounds or photoresistors if you like.



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